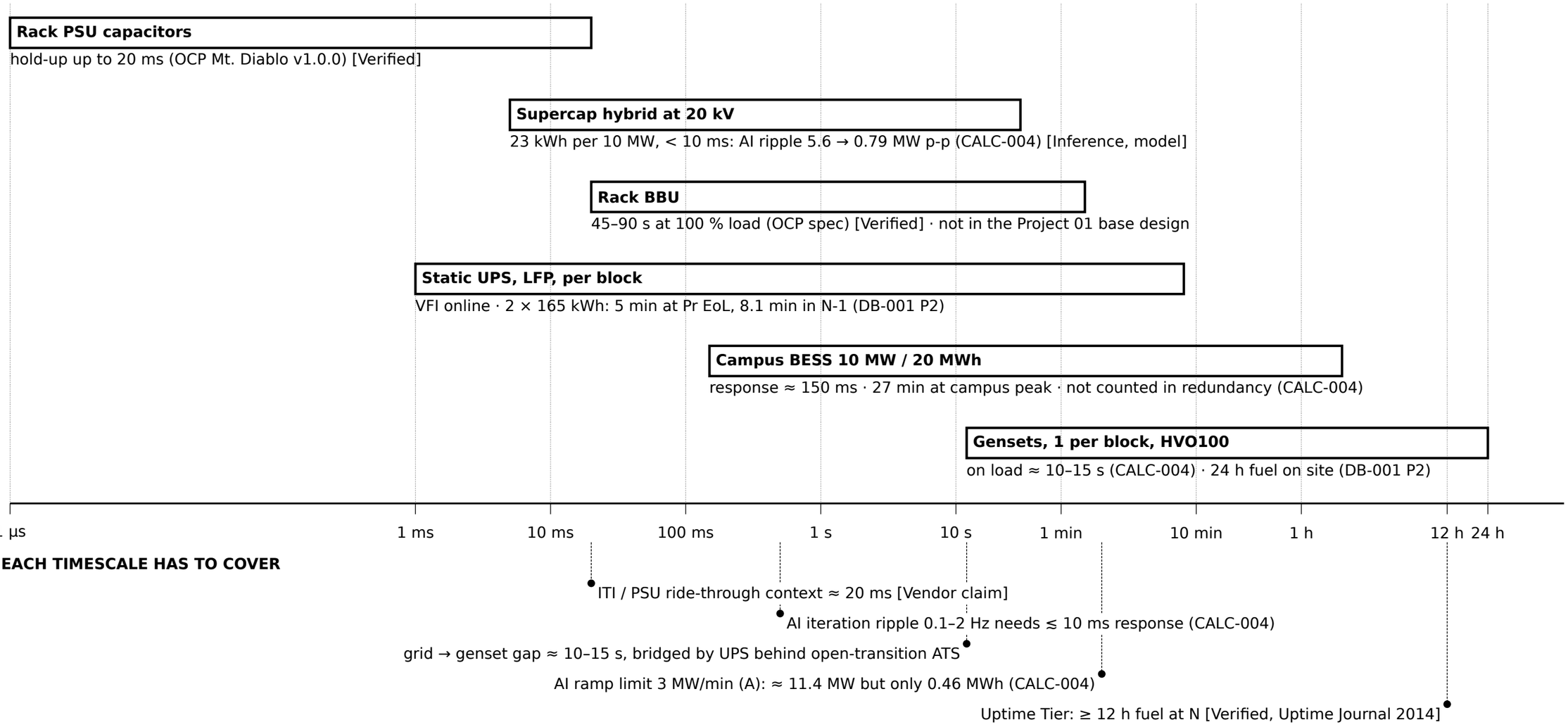
STORAGE AND SOURCE LAYERS vs TIME (campus of Projects 01 and 04)

Bar = time span the layer covers, from response time to end of energy at the stated load. Log scale.



WHAT EACH TIMESCALE HAS TO COVER

- ITI / PSU ride-through context ≈ 20 ms [Vendor claim]
- AI iteration ripple 0.1–2 Hz needs ≲ 10 ms response (CALC-004)
- grid → genset gap ≈ 10–15 s, bridged by UPS behind open-transition ATS
- AI ramp limit 3 MW/min (A): ≈ 11.4 MW but only 0.46 MWh (CALC-004)
- Uptime Tier: ≥ 12 h fuel at N [Verified, Uptime Journal 2014]

READING THE CHART

· No single layer covers 1 µs to 24 h. Fast layers have little energy; long layers respond slowly.
· The campus BESS cannot replace the UPS (150 ms response, 20 kV, behind ATS) or the gensets (27 min) — CALC-004 §4.
· AI ripple is a ms-to-s problem: capacitors / supercap, not a 20 kV BESS alone (5.6 MW p-p residual at 150 ms).
· Labels: [Verified] = source opened 2026-10-29; (A) / model values = Project 04 assumptions; [Vendor claim] not a design value.
Codes (pointers, [Unverified]): EN 50600-2-2 · IEC 62040 · ISO 8528-1 · IEC 62933-5-2.

v0.1 2026-10-29 Research draft, Project 04 context	
Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3 Portfolio D. Wu · Project 04 context (draft) · public sources, scale NTS · A3	
Storage layers vs time (draft) Context figure for Project 04 · block diagram, no ratings changed CONCEPT / DRAFT — requires review by a licensed electrical engineer	
Drawing no. **DWU-DC-E-FIG-004S**	Sheet **1 / 1** Rev **v0.1**